ALA-1

Error Code:

```
print("Student Grade System")
students = int(input("Enter number of students: "))
i = 0
total_marks = 0
pass_count = 0

while i < students:
    marks = int(input("Enter marks: "))
    total_marks = total_marks + marks

    if marks >= 40:
        pass_count = pass_count + 1
        print("Pass")

    else                                # ❌ Error 1: Missing colon (:)
        print("Fail")
        i = i + 1                       # ❌ Error 2: Inside else → may cause infinite loop

average = total_marks / students

print("Average:", average)

print("Pass count:", pass_count         # ❌ Error 3: Missing closing parenthesis )

if average >= 75:
    print("Excellent class")
else:
    print("Average class")
```

Solution:

Error 1: Missing : after else

Python requires a colon.

Correct: else:

Error 2: i = i + 1 inside else

This means:

- i increases only when student fails.
- If student passes, i does NOT increase.
- That creates an infinite loop.

Solution: Move i = i + 1 outside the if-else block but inside the while loop.

Error 3: Missing) in print

print("Pass count:", pass_count

Missing closing parenthesis.') '

Correct: print("Pass count:", pass_count)

Solved code:

```
print("Student Grade System")
students = int(input("Enter number of students: "))

i = 0
total_marks = 0
pass_count = 0

while i < students:
    marks = int(input("Enter marks: "))
    total_marks = total_marks + marks

    if marks >= 40:
        pass_count = pass_count + 1
        print("Pass")
    else:
        print("Fail")

    i = i + 1

if students > 0:
    average = total_marks / students
else:
    average = 0

print("Average:", average)
print("Pass count:", pass_count)

if average >= 75:
    print("Excellent class")
else:
    print("Average class")
```

output:

```
Student Grade System
Enter number of students: 5
Enter marks: 99
Pass
Enter marks: 88
Pass
Enter marks: 77
Pass
Enter marks: 66
Pass
Enter marks: 55
Pass
Average: 77.0
Pass count: 5
Excellent class
```